(ii) Find the instantaneous centre of link AB relative to ground at the instant of time given in (i) above. [7marks]

(iii) Hence or otherwise find the angular velocity of link AB, ω_{31}, [3marks]

Q3. (a) Prove that the maximum velocity of a vehicle of track B moving on a horizontal road in a bend of R radius to prevent overturning and skidding are given by:

(i) $V_{overturn} = \sqrt{\left(\frac{BRg}{2h}\right)}$ (ii) $V_{skid} = \sqrt{(\mu R g)}$ [10 Marks]

(Note: All symbols have their usual meaning)

(b) The wheel track of a motor van is 3.2m and its centre of gravity is 0.78m above the road.

(i) Calculate the speed at which the van will overturn travelling around a bend of 40m radius assuming the road is level. [5marks]

(ii) Calculate the speed at which the van will skid travelling under the same condition in (i) above. Take the coefficient of friction between road and tyres as 0.35. [5Marks]

(iii) Which of the two conditions in (i) and (ii) will never occur? Explain [2 Marks]

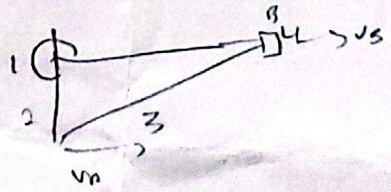
(iv) If the horizontal road above is to be banked to an angle of **30°** to the horizontal. Draw the free body diagram and hence or otherwise calculate the maximum velocity of the van to prevent skidding when the bend is now 30m radius. [8 Marks]

Q4. (a) Given that $P = \frac{F}{2\pi r(R_o - R_i)}$ using the uniform wear theory show from first principle that the torque Transmitted by a friction clutch is given as $T = \mu F R_m n$ [10 Marks]

Hint: $T = \int_{R_i}^{R_o} 2\mu\pi P r^2 \, dr$: All symbols used have their usual meaning

(b) A multiple plate clutch having four active surfaces transmits 40 kW of power and the maximum torque developed is 160Nm. Axial pressure is not to exceed 120kN/m². Outer diameter of the friction plate is 1.5 times the inner diameter. Determine these diameters and the axial force exerted by the springs. Assume uniform wear and take coefficient of friction as 0.30.

[20 Marks]



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